Result 1 (finite family application). [Topology foundations, dependent proof family application]
Let n be a natural number. Let $f = \mathrm{refl}_n$. Let v be a vertex in $\{0, \ldots, n-1\}$. Let $h = f(v)$. Then

$$v = v.$$

Result 2 (empty family). [Topology foundations, proof family over Fin zero]
Let f be a function assigning to each $v : \{0, \ldots, 0-1\}$ evidence that $v = v$. Let v be a vertex in $\{0, \ldots, 0-1\}$. Let $h = f(v)$. Then

$$v = v.$$

Result 3 (singleton family). [Topology foundations, proof family over Fin one]
Let f be a function assigning to each $v : \{0, \ldots, 1-1\}$ evidence that $v = v$. Let v be a vertex in $\{0, \ldots, 1-1\}$. Let $h = f(v)$. Then

$$v = v.$$

Result 4 (proof domain application). [Topology foundations, explicit proof-domain function]
Let n be a natural number. Assume $n = n$ (`same`). Let $f = \mathrm{id}$. Let $h = f(\mathit{same})$. Then

$$n = n.$$

Result 5 (module family application). [Topology foundations, higher-universe proof family application]
Let K be a type. Let *field* be a field structure on K. Let M be a module over K. Let x be an element of M. Let $f = \mathrm{refl}_K$. Let $g = f(M)$. Let $h = g(x)$. Then

$$x = x.$$

LLM-assisted example and proof (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.